# HISTORY OF CEMENT AND CONCRETE IN INDIA – A PARADIGM SHIFT

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## Abstract

Since independence in the year 1947, cement in India remained a controlled commodity for nearly four decades. The price and distribution of cement were controlled by the Government. In the controlled era, prices fixed were not attractive enough to reinvest in the cement industry, to modernize plant and machinery and to create new capacities; hence there was a gap between demand and supply and cement shortages continued all along the control period. As a result, both, the cement industry and the consumers suffered. Cement produced was as per the specifications laid down by the Bureau of Indian Standards (BIS). The choice of cement (and concrete) as per the application was unknown and the situation did not offer any motivation towards productivity enhancement of cement manufacturing units as well as towards the quality improvement of cement. As a part of liberalization policy, the Government of India started gradual "decontrol" of cement in 1977 which then gave 12% post tax return on net worth on new capacity creation. It provided the initial boost to cement industry followed by partial decontrol in 1982 and finally cement became a 'free commodity' in the year 1989. These policy decisions made a positive impact both on the quality as well as the economy of the cement industry. This has helped in reversing the situation of cement shortages to cement surplus. Cement is now available off-the-shelf. The cement industry has made phenomenal progress in terms of volume, technology and product upgradation. It has the state-of-the-art modern large capacity plants. The quality of Indian cement is at par with the best produced anywhere in the world. Today India is world's second largest cement producing country, with an installed capacity of 160.24 x $10^6$ t/a. The energy consumption too - both thermal and electrical - per unit mass production was brought down, through productivity enhancement and modernization efforts on the part of cement plants. The quality of cement also improved impressively, with better strength and durability characteristics. The most significant aspect of this change is the shift in consumer preference from 'high strength' to 'high performance' or durable concrete, which was prompted by greater quality awareness among the consumers and high cost of structural repairs. This general improvement in the quality of concrete was achieved through greater use of mineral admixtures, which are industrial wastes, namely fly ash and blast furnace slag, in cement and concrete. The national standards and the codes of practice have also been revised and supported the change. The greater use of industrial wastes as mineral admixtures has led the cement and construction industry in India on the path of sustainable development.

## Introduction

The Indian cement industry is today nearly 93 years old. The first plant was established in 1913 with an annual capacity of 0.01 x $10^6$ t/a. The installed capacity in 2006 stood at 160.24 x $10^6$ t/a. The growth of cement industry in India has seen many a boom and slack period; the most significant being the shift from the "control regime" to "free regime" in the year 1989. The cement and concrete industry witnessed both quantitative as well as qualitative growth in the free regime, which continues till date. The paper reviews history of this growth in three sections namely, (1) Cement industry: growth from control to free regime, (2) Energy-materials-intensive to efficient and sustainable cement manufacture, (3) Preference of construction industry from high strength to durable, high performance concrete

## 1. Cement industry: growth from control to free regime:

The foundation of cement industry in India was laid by India cement company Ltd. in the year 1912-13 at Porbundar (Gujarat State) and started production in 1914. The annual capacity of cement production then was 0.01 x $10^6$ t. The production of cement gradually increased to 4.6 x $10^6$ t/a, during the first five year plan (1951-1956)[1,2] of the Government of India.

The price and distribution of cement, from 1 July 1956, came under the provisions of control order promulgated in the exercise of the powers conferred under section 18 (G) of the Industries Act 1951[2]. The State Trading Corporation (STC) was established for the acquisition and distribution of cement. The main objective was to ensure that both the indigenously manufactured and the imported cement is sold at the same price. The STC was required to rationalize the movement of cement. The prices payable to cement producers were fixed on ex-works basis for naked cement. After adding the seller's commission, average freight, packing charges, excise duty and other charges, a uniform free-on-road (F.O.R) price was fixed, at which STC was required to sell the cement all over the country. The prices of the cement were revised time-to-time with regard to the basic factors viz. excise duty, freight and packing charges, in consultation with the various Government Departments. The price and distribution of cement continued in the controlled regime till 1977, when as a part of liberalization policy, the Government of India initiated gradual decontrol by giving 12% post tax return on net worth on new capacity creation, followed by partial decontrol in 1982. The controlled price did not reflect the economic cost. Thus continuous lower level of controlled prices acted as a disincentive for productivity which resulted in increasing shortages of cement. In order to improve the overall situation, the cement industry was finally put on the path of "free regime" by introducing total decontrol of price and distribution in March 1989. The cement industry registered a significant growth in the post decontrol period.

The Table 1[1,2,3,4] gives the growth of installed capacity and production of cement for past five decades. As can be seen from the Table, the installed capacity of cement plants increased from 5.99 x $10^6$ t/a in 1956 to 42.35 x $10^6$ t/a in three decades i.e. till the year 1986. As a result of free regime, the installed capacity registered phenomenal growth in the last two decades. The annual installed capacity of nearly 118 x $10^6$ t was added during 1986 to 2006.

**Table 1 Installed capacity and production of cement in India (1956 – 2006)** [1,2,3,4]

| Sr.No. | Year | Installed Capacity ($10^6$ t) | Production ( $10^6$ t) | | | | |
|---|---|---|---|---|---|---|---|
| | | | OPC | PPC | PSC | Others | Total |
| 1 | 1956 | 5.99 | NA | NA | NA | NA | 5.02 |
| 1 | 1960 | 8.97 | NA | NA | NA | NA | 7.80 |
| 2 | 1966 | 11.69 | NA | NA | NA | NA | 11.06 |
| 3 | 1970 | 19.80 | NA | NA | NA | NA | 14.27 |
| 4 | 1976 | 21.33 | NA | NA | NA | NA | 18.61 |
| 5 | 1980 | 25.52 | 6.08 (32.8) | 9.35 (50.4) | 3.01 (16.2) | 0.11 (0.6) | 18.56 |
| 6 | 1986 | 42.35 | 13.05 (40.7) | 14.44 (45.1) | 4.36 (13.6) | 0.20 (0.6) | 32.05 |
| 7 | 1990 | 56.96 | 30.59 (71.3) | 7.45 (17.4) | 4.62 (10.8) | 0.25 (0.6) | 42.91 |
| 8 | 1996 | 88.25 | 45.04 (69.8) | 11.77 (18.2) | 7.10 (11.0) | 0.62 (0.96) | 64.53 |
| 9 | 2000 | 119.30 | 62.76 (66.62) | 21.30 (22.61) | 9.39 (9.97) | 0.76 (0.8) | 94.21 |
| 10 | 2001 | 133.41 | 58.06 (62.02) | 24.50 (26.17) | 10.34 (11.05) | 0.71 (0.76) | 93.61 |
| 11 | 2002 | 137.86 | 57.68 (56.32) | 32.30 (31.54) | 11.89 (11.61 | 0.53 (0.53) | 102.40 |
| 12 | 2003 | 145.18 | 56.05 (50.34) | 43.09 (38.70) | 11.63 (10.44) | 0.58 (0.52) | 111.35 |
| 13 | 2004 | 153.85 | 53.51 (45.54) | 52.13 (44.37) | 11.26 (9.58) | 0.60 (0.51) | 117.50 |
| 14 | 2005 | 159.80 | 55.97 (43.87) | 60.24 (47.22) | 10.73 (8.14) | 0.63 (0.5) | 127.57 |
| 15 | 2006 | 160.24 | 55.84 (39.38) | 74.01 (52.19) | 11.37 (8.02) | 0.59 (0.41) | 141.81 |

*Notes : (a)* Others *include oil-well, low heat, special cements, sulphate resisting cement and others*

*(b) OPC – Ordinary Portland Cement, PPC – Portland Pozzolana Cement PSC – Portland Slag Cement*

*(c) Figures in bracket indicate percent of total production.*

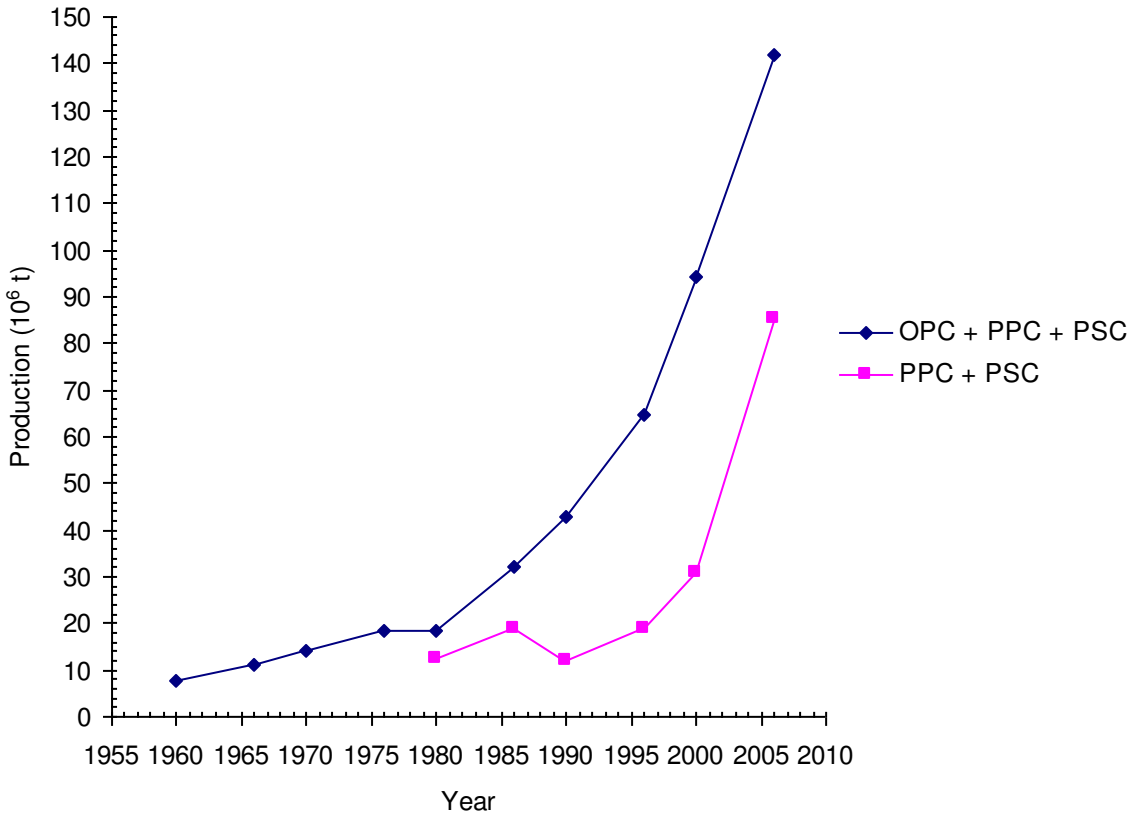


**Fig.1 Trends of cement production in India (1956-2006)**

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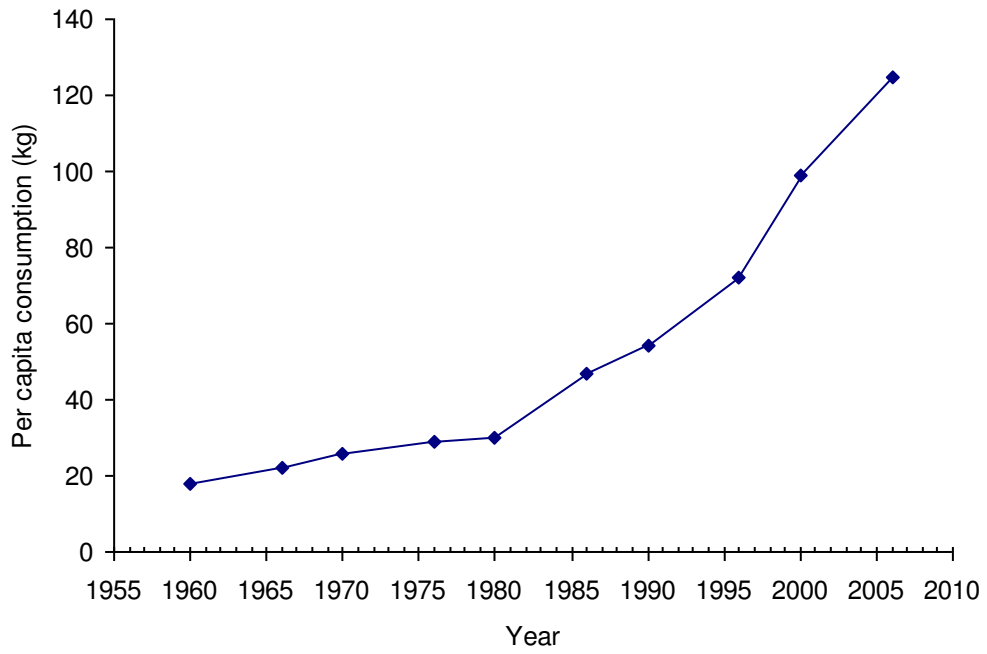


**Fig.2 Growth of per capita consumption of cement in India (1956-2006)[1,2,3,4]**

The trends are visible in Fig.1. The Figure also shows the growth in the manufacturing capacity of blended cement, based mainly on utilization of industrial wastes, namely fly ash (FA) and blast furnace slag (BFS).

The Fig. 2 shows the growth of per capita cement consumption[1,2,3,4]. As can be seen from the Figure, after registering an almost negligible growth during the control regime, the per capita cement consumption rose almost exponentially after the introduction of partial decontrol in the year 1982, followed by full decontrol in the year 1989

The Fig.3. shows the trend of profitability in terms of post tax returns of cement industry upto 1997[5]. As can be seen in the Figure, after registering a negative growth in control and partial control regime, the profitability of cement industry started looking up in the free regime.

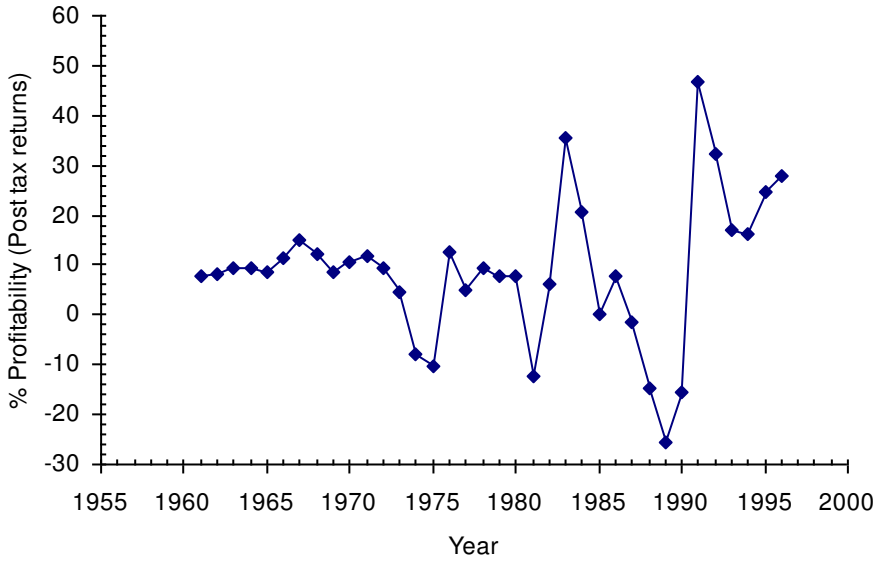


**Fig.3 Profitability trend of cement industry in India (1956-1997)[5]**

The improvement in the profitability continues. This aspect is illustrated in the Fig.4 which shows the combined net profit of selected 23 Indian cement companies, since the year 2004[6,7,8].

As a result of overall growth in the capacity generation, the cement industry of India started making a mark in the global scenario. The Table 2 shows the world ranking of Indian cement industry. As can be seen from the Table, the global ranking of Indian cement industry in forms of annual production improved from fourth position in 1993 to second in 2002, next to China, which still holds. It should be noted that despite the growth, the per capita cement consumption in India has remained lower than the world average per capita consumption. As per the available information the world average per capita cement consumption in the year 2003 was 307 kg/a as against that of 110 kg/a in India, which leaves a substantial scope for further growth of the cement industry[5].

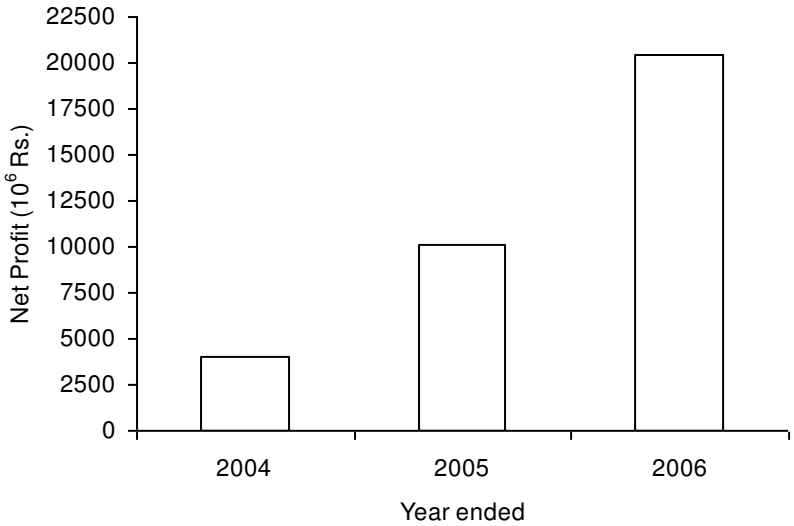


**Fig.4 Combined net profit of selected 23 Indian cement companies[6,7,8]**

**Table 2 World ranking of Indian cement industry [2,3,9]**

| Sr. No. | Country | Production ($10^6$ t) in the year | | |
|---|---|---|---|---|
| | | 1993 | 1997 | 2002 |
| 1 | China | 360 | 514.3 | 704.14 |
| 2 | Japan | 94.8 | 95.8 | 72.13 |
| 3 | USA | 73.3 | 82.6 | 89.73 |
| 4 | India | 58.5 | 82.6 | 117.52 |
| 5 | Other countries | 706.4 | 740.1 | 834.40 |

## 2. Energy-materials-intensive to efficient and sustainable cement manufacturing:

As discussed in Section 1, the transition of cement industry from control to free regime gave fillip to growth. The improved profitability resulted in the addition of installed capacity. It went hand-in-hand with the improvement in the manufacturing technology and consequent savings in the energy, conservation of raw materials and fuels, leading to sustainable growth of the cement industry. The Table 3[10] shows the change in the process profile over the years. As can be seen from the Table, the share of wet, semi-wet/dry process came down from 100% to 4% of the total number of kilns.

**Table 3 Changing process profile of Indian cement industry[10]**

| Sr. No. | Particulars | Year | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | 1950 | 1960 | 1970 | 1983 | 1995 | 2001 | 2006 |
| 1 | Wet, Semi-wet/dry Process | | | | | | | |
| | No. of Kilns | 33 | 73 | 101 | 104 | 69 | 40 | 34 |
| | % of the total | 100 | 98.9 | 78.5 | 46.8 | 14 | 7 | 4 |
| 2 | Dry Process | | | | | | | |
| | No. of Kilns | --- | 1 | 18 | 50 | 97 | 117 | 128 |
| | % of the total | --- | 1.1 | 21.5 | 53.2 | 86 | 93 | 96 |

The Indian cement industry today employs the most modern manufacturing technology in terms of the unit operations. Some of the modern features of Indian cement manufacturing are given as follows:

- Computerised mine planning
- Efficient blending systems
- Energy efficient comminution (size reduction) processes namely vertical roller mills and roller press
- High efficiency air separators
- Improved dust collection systems employing fabric filters and electrostatic precipitators
- Suspension pre-heaters and pre-calciners
- High capacity kilns with improved heat transfer and low energy requirement
- High efficiency clinker coolers
- Energy efficient and less polluting materials handling systems
- Modern high capacity, electronically controlled bag/bulk packing and dispatch systems
- Moisture and seepage resistant cement packaging

The Table 4[11] shows the average power consumption in different sections of cement manufacture.

**Table 4 Average specific electrical energy consumption in cement manufacture in India[2,4,11]**

| Section | Electrical energy, kWh/t (Average) | |
|---|---|---|
| Mining and crushing | 1.5 | |
| Raw material grinding (dry)<br>Ball mill<br>Vertical roller mill | <br>22<br>17 | |
| Coal grinding<br>Ball mill<br>Vertical roller mill | <br>30<br>25 | |
| Kiln section: pyro processing<br>Dry PC<br>Dry SP | <br>26<br>28 | |
| Finish grinding | Ball mill (closed circuit) | Roller press/VRM (finish mode) |
| OPC-33[a] | 32 | 28 |
| PPC[b] | 32 | 28 |
| PSC[c] | 50 | 28 |
| Packing and Despatch | 2 | |

*Notes: (a) Blaine fineness about 280 $m^2/kg$*
*(b) Blaine fineness about 320 $m^2/kg$*
*(c) Depends on slag percent and its quality*

As can be seen from Table 4, the reduction in power consumption has been mainly brought about by the use of vertical roller mill and roller press, which replace ball mill in the comminution circuits of raw materials grinding, coal grinding and finish (cement) grinding. In addition, the improved energy efficient equipment in materials handling brought about the overall reduction in the average power consumption in cement manufacture, as can be seen in Table 5[12].

**Table 5 Trend of energy consumption of cement industry in India[3,12]**

| Sr. No. | Year | Energy consumption (Average Kcal/kg clinker) | Power consumption (Average kWh/t cement) |
|---|---|---|---|
| 1 | 1999-2000 | 748 | 91 |
| 2 | 2000-2001 | 740 | 88 |
| 3 | 2001-2002 | 742 | 87 |
| 4 | 2002-2003 | 734 | 84 |
| 5 | 2003-2004 | 732 | 83 |
| 6 | 2004-2005 | 723 | 82 |
| 7 | 2005-2006 | 720 | 82 |

The Table 5[3,12] also shows a gradual reduction in the average thermal energy consumption that has been brought about by improved coal burners and reduction in the energy losses through overall productivity improvement efforts. The Table 6[13] shows the comparison of best average energy consumption levels achieved by the state-of-the-art plants in Japan, India, South Korea, Taiwan and Thailand.

**Table 6 Best average energy consumption level of the state-of-the-art cement plants in some Asian countries (1997-98 data)[13]**

| Best average | Japan | India | South Korea | Taiwan | Thailand |
|---|---|---|---|---|---|
| Power (kWh/t cement) | 97.8 | 80-85 | 99-100 | 98-100 | 103-105 |
| Fuel (kcal/kg clinker) | 653 | 690 | 710 | 730 | 720 |

As seen in Table-1 and Fig.1, the production of blended cement also increased in the growth period and crossed the 50% mark in the year 2004, making blended cement as the major variety of cement manufactured in India. The blended cement is manufactured by replacing Portland cement by pozzolanic materials such as FA or cementitious materials like BFS. The percentage of cement replacement by FA and BFS is stipulated by the respective national standards. The Indian specification for Portland pozzolana cement[14] allows replacement of cement upto 25-35% by FA and the specification of Portland slag cement[15] allows replacement of cement upto 25-70% by BFS. The manufacture of blended cement results in substantial saving in thermal and electrical energy consumption, as can be seen in the comparative evaluation given in Table 7[16] .

The Table 8 shows the generation of major industrial wastes and their utilization in cement. The Table highlights that there is still a good scope for greater utilization of industrial wastes in cement manufacture.

**Table 7 Thermal and electrical energy consumption in blended cement manufacture[16]**

| Sr.No. | Cement | Energy consumption ( existing plant) (%) | |
|---|---|---|---|
| | | Thermal | Electrical |
| 1 | Ordinary Portland cement | 100 | 100 |
| 2 | Portland pozzolana cement[a] | 84 | 89 |
| 3 | Portland slag cement[b] | 47 | 83[c] |

Notes: (a) *Assuming 15% pozzolana addition*
*(b) Assuming 50% slag addition*
*(c) Average for old and new plants*

**Table 8 Generation of major industrial wastes and their utilization in cement in India**

| Waste material | Generation ($10^6$ t/a) | Utilization ($10^6$ t/a) |
|---|---|---|
| Fly ash | 125 | 19 |
| Blast furnace slag | 10 | 5.95 |

The manufacture of cement requires a substantial quantity of limestone, clay as raw materials and coal as a fuel. These are non-renewable natural resources. The manufacture of blended cement results in saving of the natural resources as can be seen in Table 9.

**Table 9 Saving of natural resources due to manufacture of blended cements (Basis: One tonne cement)**

| Cement type | Raw materials[a] quantity (t) | Coal quantity (t) |
|---|---|---|
| OPC | 2 | 0.18 to 0.20 |
| PPC[b] | 1.7 | 0.15 to 0.17 |
| PSC[c] | 1 | 0.09 to 0.1 |

*(a) Limestone, clay and additives. (b) Assuming 15% cement replaced by flyash (c) Assuming 50% cement replaced by blast furnace slag*

The overall conservation of raw materials, mineral coal and efficient burning and heat treatment techniques in cement plants have resulted in the gradual reduction of emission of carbon dioxide (green house gas), as can be seen from the Table 10[10,17].

**Table 10 Trend of $CO_2$ emission due to cement manufacture in India[10,17]**

| Year | $CO_2$ emission (t/t of cement produced) |
|---|---|
| 1995-96 | 1.192 |
| 1996-97 | 1.186 |
| 1997-98 | 1.086 |
| 1998-99 | 0.994 |
| 1999-00 | 0.975 |
| 2000-01 | 0.980 |
| 2001-02 | 0.954 |
| 2002-03 | 0.90 |

Table-11[13] shows the comparison of $CO_2$ emission per tonne of cement produced in Japan, USA, China and India. As can be seen in the Table, there is still a substantial scope for reduction in $CO_2$ emission in Indian cement plants, through greater use of industrial wastes.

**Table 11 Comparison of $CO_2$ emission of major cement producers in the world (2003 data)[13]**

| Country | $CO_2$ emission (t/t of cement produced) |
|---|---|
| Japan | 0.45 |
| USA | 0.47 |
| China | 0.43 |
| India | 0.9 |

## 3. Preference of construction industry from high strength to durable, high performance concrete

As seen in the preceding discussion, the approach of the Indian cement industry in the last two decades changed towards savings in energy and conservation of raw materials. As a result of high cost of repairs in the newly constructed buildings, mostly built using high strength cement, the construction industry in India appreciated the need to use durable, high performance concrete instead of high strength concrete. The following developments in concrete industry highlight that aspect[2,4,5].

a) Reduction in water-to-cement ratio using chemical admixtures, leading to the improvement in strength and durability of concrete.

b) Use of lesser cement, leading to conservation of building materials.
c) Increased use of mineral admixtures such as FA and BFS, leading to greater utilization of industrial wastes.
d) Greater application of ready mix concrete instead of site-mixed concrete, leading to overall improvement in the concrete quality.

The change in approach of the Indian construction industry was also reflected in the revision of the Indian code of practice for "Plain and Reinforced Concrete", I.S 456-2000[18]. The Code is commonly referred by all engineers and practitioners of Civil Engineering in India. The revised code gives special consideration to the durability of concrete structures[18,19,20]. The changes in the revised Indian Standard Code in the year 2000, as against the year 1978[19], are discussed below:

a) The workability of concrete has been expressed in terms of slump only, unlike the earlier Code, where it was expressed in terms of Vee-bee time/ compacting factor.

b) The maximum water-cement ratio for reinforced concrete has been reduced from 0.6 to 0.55, for M20 concrete.

c) The permeability of concrete to the ingress of deleterious agents has been identified as one of major characteristics affecting the durability. The factors influencing the durability have been delineated as environment, cover to embedded steel, type and quality of construction materials, cement content and water-to-cement ratio of the concrete, workmanship to obtain full compaction and efficient curing and shape and size of the member.

d)The general environment, to which the concrete is exposed during its working life, is classified into five levels of severity, namely mild, moderate, severe, very severe and extreme. The conditions of very severe and extreme have been added in the revision, with less cement content.

e) The air-entraining admixtures have been recommended for use in the concrete, where freezing and thawing actions under wet conditions exist

f) Recommendations have been given for the type of cement, maximum free water-to-cement ratio and minimum cement content, to develop adequate resistance in the concrete exposed to different sulphate concentrations.

g) The upper limit of the cement content, not including FA and BFS, has been kept at 450 kg/m$^3$.

h) The maximum chloride content (as Cl) in the concrete, at the time of placing, has been specified at 0.6 kg/m$^3$ of concrete.

i) The revised Code stipulates the maximum sulphate content of the concrete mix, expressed as $SO_3$ at 4% by mass of cement in the mix.

j) The Code has stated precautions against alkali-aggregate reaction and recommends use of non-reactive aggregates, low alkali cement and also recommends partial replacement of cement with mineral admixtures.

k) The use of ready-mixed concrete or concrete from on/off site batching and mixing plant has been recommended for large and medium size projects.

l) The Code stipulates that the concrete mixers shall be provided with water measuring devices. The provision will go a long way in controlling water-to-cement ratio, especially in site-mixed concrete.

m) The use of proper and adequate curing techniques has been stressed, to reduce the permeability of the concrete and enhance its durability by extending the hydration of cement. The difference has been maintained between recommended minimum period of curing concrete containing OPC and that containing cement with mineral admixtures.

While going through these provisions, it becomes clear that besides improving the construction quality, concrete strength and durability, the revised Code puts emphasis on conservation of building materials also.

The application of concrete in new (or less common) areas as well as the use of industrial wastes, such as FA or BFS, in concrete is increasing. The following areas appear to have great potential under Indian conditions:

**Concrete roads:** The total length of roads in India, as in March 2002, was nearly 2.5 x $10^6$ km. Out of those, nearly 58 % are surfaced roads. The share of cement concrete roads, within surfaced roads, is less than 2 %. According to one study conducted jointly by the Ministry of Rural Development and the Ministry of Commerce, Government of India, the life-cycle cost of concrete roads works out to be 5.7 % less than that of the bituminous roads, the higher initial investment notwithstanding. The use of concrete in road construction is increasing gradually. The Plate 1 shows a section of 95 km Mumbai-Pune expressway constructed in concrete recently.

It is important to build the rural roads with concrete, considering the fact that they often remain neglected from the regular maintenance. The Plate 2 shows a typical rural road constructed in concrete. It will be worthwhile mentioning statistics in some Asian countries, for the sake of comparison. In Japan, nearly 10.5 % of the cement produced goes in the construction of roads and bridges, whereas in China the infrastructure sector, which includes roads, consumes about 40% of the cement. Thus there is a huge potential for the consumption of cement in road construction in India, provided the long-term superiority of concrete roads over bitumen roads is appreciated at all levels[21]

**Plate 1 : Mumbai-Pune expressway (India) in concrete** **Plate 2: Rural road in concrete**

*(Courtesy: Cement manufacturers' Association, India)*

**High volume fly ash concrete (HVFAC):** The use of high volume fly ash concrete (HVFAC) finds greater acceptability in the Indian constructions[22,23,24,25,26,27]. The HVFAC aims at replacing cement with FA in percentages more than 50 (mass/mass). Thus the larger application of HVFAC in construction leads to greater utilization of FA. The Plate 3 shows section of Ghatghar dam (Maharashtra State)

constructed with roller compacted concrete using 60% fly ash, as replacement of cement.

**Plate 3 : Ghatghar dam in roller compacted concrete with 60% fly ash**
*(Courtesy: Government of Maharashtra, India)*

The Table 12[28] shows the mix design of the roller compacted concrete used in the construction of Ghatghar dam in India

**Table 12 Mix design for M15 (91 days) concrete for Ghatghar Dam project[28]**

| Cement Kg/m$^3$ | Fly ash Kg/m$^3$ | Water Kg/m$^3$ | Sand Kg/m$^3$ | Coarse aggregate, kg/m$^3$ | | | Admixtures kg/m$^3$ |
|---|---|---|---|---|---|---|---|
| | | | | 50-20 mm | 20-10 mm | 10-5 mm | |
| 88 | 132 | 60 | 692 | 558 | 446 | 406 | 2.2 |

Table 13[29] shows another example of the use of fly ash in high volumes in concrete. It gives the mix design for M30 Grade self compacting concrete (SCC) with 50% replacement of cement with flyash, used at Kaiga atomic power project (India)

**Table 13 Mix proportions for M30 Grade concrete used at Kaiga atomic power project[29]**

| Cement (kg/m$^3$) | Fly ash (kg/m$^3$) | Water (kg/m$^3$) | Coarse aggregate (kg/m$^3$) | Sand (kg/m$^3$) | Admixtures (kg/m$^3$) | Viscosity modifying agent (kg/m$^3$) | Average compressive strength (MPa) |
|---|---|---|---|---|---|---|---|
| 225 | 225 | 165 | 713 | 978 | 5.17 | 1.35 | 39.6 |

**Use of pond ash in construction:** As mentioned earlier (Table 8 ), only about 15% of the total FA generated in India, mostly collected in dry condition from electrostatic precipitators in thermal power plants, is presently utilized in the cement and concrete. The remaining FA and bottom ash is disposed of in open ponds/lagoons, creating huge land, water and air pollution. The ash disposed of in lagoons occupies an area of more than 50000 ha. The effort towards large-scale utilization of this ash

is welcome and will go a long way in the sustainable development of the thermal power generation industry. One of the effective ways to utilise this waste material, on a large scale, is to add it in concrete in high volumes as a partial replacement of Portland cement (PC), thereby contributing towards the sustainable development of cement and construction industry also. According to a recent research finding reported by Bapat J. D., et.al.[30], the pond ash or lagoon ash can be successfully used in dry lean concrete, a sub-base for concrete roads. In the laboratory experiment, 55% cement was replaced with lagoon ash to make dry lean concrete. In another research report, Prasad K. S. S. V. V.[31] reports successful use of pond ash in the manufacture of bricks. The use of pond ash in dry lean concrete or in bricks opens the prospects of using the industrial waste on a large scale in the construction activity.

## Conclusions

1. The "control regime" imposed on Indian cement industry for the period 1956-1989 offered non-remunerative prices for cement, hence stunted the overall growth of cement industry. The "free regime", introduced since the year 1989, improved profitability and resulted in the addition of installed capacity on the one hand and encouraged cement consumption in different sectors of economy, on the other.
2. The addition of installed capacity went hand-in-hand with the improvement in manufacturing technology and consequent savings in the energy, conservation of raw materials and fuels and greater utilization of industrial wastes in cement manufacture, leading to sustainable growth of the industry.
3. The preference of Indian construction industry has today changed from high strength to durable and high performance concrete. As a result of the new approach, concrete mix is engineered to make appropriate use of chemical admixtures and mineral admixtures. The revised national Code for plain and reinforced concrete has also made necessary provisions. The current trend is to use fly ash in high volumes in concrete and application of concrete in relatively less common areas such as concrete roads.

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